Object DSL

Shape

Rigid

Soft

Physical

Visual

Entity DSL

Falling Ball

Soft Pad

Objects List

Example: falling_objects_on_soft_pad

Objects Number N

Height H_1

Position (X_1, Y_1)

Entity–Specific Parameters

Scene DSL

Entity

HDRI

Camera

Cross-Material Simulation References

Example: dominoes

Example: draining_tank

Physical Taxonomy

Coupling

fluid-solid interaction

gas-solid interaction

soft-rigid interaction

Gas

vortex dynamics

combustion

dispersion & mixing

buoyancy

thermal convection

advection

Rigid

collision

momentum conservation

gravity

friction

resting stability

Liquid

free-surface evolution

pressure

mass conservation

buoyancy

advection

Soft

bending

inextensibility

draping & wrinkling

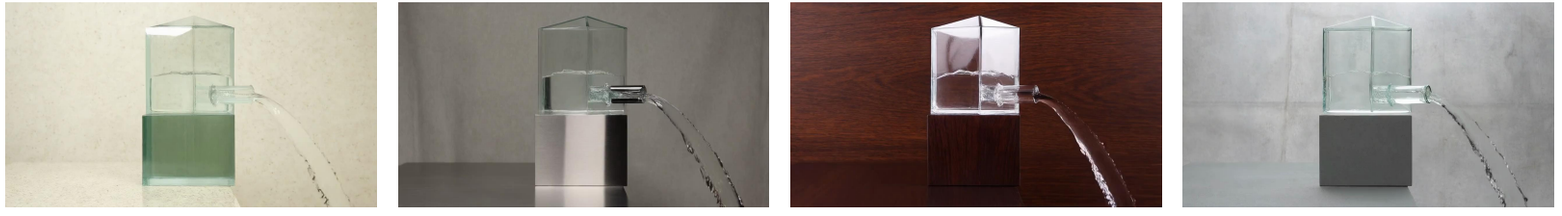
elastic recovery

damping dissipation

Example: flame_heating

Example: falling_objects_on_soft_pad

Photorealistic Translation



Behavior-Specific Metric Evaluation